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Chapter 1: Governing Equations in Mantle Dynamics

1.1 Mechanics in Mantle Dynamics

Extensive studies have been conducted on fluid dynamics of a two-phase mixture. We briefly discuss the four governing equations of mechanics, drawing on a similar discussion presented by Drew and Segel (1971), McKenzie (1984) and Katz (2022).

1.1.1 Phase Fraction

We begin with defining the phase fraction using an indicator function, under the assumption that multiple phases cannot coexist at the same spatial location. Accordingly, the phases are considered immiscible. The indicator function is defined as

$$\chi_i(\mathbf{x}) = \begin{cases} 1 & \text{phase } i, \\ 0 & \text{not phase } i. \end{cases} \quad (1.1)$$

We then define the macroscopic phase fraction function of phase i within a representative volume element V as

$$\phi_i = \frac{1}{V} \int_V \chi_i(\mathbf{x}) d\mathbf{v}, \quad (1.2)$$

where $\sum_i \phi_i = 1$. In the incompressible setting, the phase fraction may be equivalently regarded as the volume fraction occupied by each phase within a representative elementary volume, reflecting the proportion of space occupied by each phase at the macroscopic scale. Phase-averaged (or volume-averaged) quantities, also referred to as bulk quantities, are defined as

$$\bar{\gamma} = \sum_i \phi_i \gamma_i, \quad (1.3)$$

where γ_i denotes a quantity that may be scalar- or vector-valued.

1.1.2 Conservation Laws

We consider the conservation equation for a single-phase system. For a given volumetric quantity γ defined within a representative volume element V , we can write a corresponding conservation equation as:

$$\frac{d}{dt} \int_V \gamma d\mathbf{v} = - \int_{\partial V} \mathbf{f} \cdot d\boldsymbol{\sigma} + \int_V S d\mathbf{v}, \quad (1.4)$$

where $\mathbf{f}$ represents flux travelling across the boundary of a representative volume element V . The minus sign indicates that the positive direction aligns with normal unit vector pointing outwards. The source/sink term S , represents the volumetric production rate of the volumetric quantity γ . We apply Gauss's theorem to obtain the differential form

$$\frac{\partial \gamma}{\partial t} + \nabla \cdot \mathbf{f} = S. \quad (1.5)$$

We model the partially molten mantle as a two-phase mixture consisting of liquid magma and solid mantle. The phase fraction of liquid magma is denoted by ϕ_l and phase fraction of solid mantle is labelled as ϕ_s , such that the relation $\phi_l + \phi_s = 1$ is always obtained.

We rewrite the single-phase conservation law (1.5) with the bulk density as

$$\frac{\partial \bar{\rho}}{\partial t} + \nabla \cdot \bar{\rho} \bar{\mathbf{v}} = 0, \quad (1.6)$$

where $\bar{\rho} = \phi_l \rho_l + \phi_s \rho_s$. Under the Boussinesq approximation, we neglect the compressibility of each individual phase. Furthermore, we extend the Boussinesq approximation by disregarding the density difference between the two phases, except in the body force term. Accordingly, the continuity equation for the two-phase mixture can be written as

$$\nabla \cdot \bar{\mathbf{v}} = 0. \quad (1.7)$$

Similarly, the conservation equation for the phase-averaged momentum takes the form

$$\frac{\partial \bar{\rho} \bar{\mathbf{v}}}{\partial t} + \nabla \cdot \overline{\rho \mathbf{v} \otimes \mathbf{v}} = \bar{\rho} \mathbf{g} + \nabla \cdot \bar{\boldsymbol{\tau}} - \nabla \bar{p}, \quad (1.8)$$

where $\boldsymbol{\tau}$ represents the deviatoric stress. By neglecting inertial effects, justified by the low Reynolds number of magma and mantle, the static force balance equation reduces to

$$\bar{\rho}\mathbf{g} + \nabla \cdot \bar{\boldsymbol{\sigma}} = 0, \quad (1.9)$$

where $\boldsymbol{\sigma} = \boldsymbol{\tau} - p\mathbf{I}$ represents the total stress, consisting of the deviatoric component $\boldsymbol{\tau}$ and the isotropic pressure $p\mathbf{I}$.

1.1.3 Darcy's Law

The equations governing bulk quantities (1.7) and (1.9) alone are insufficient to simultaneously determine the two distinct velocities and pressures. Therefore, an additional constitutive relationship is required to describe the behavior of the liquid phase within the porous medium. We examine more closely the interphase forces between the solid and liquid phases. A brief overview of the derivation of the individual governing equations for each phase is provided here, while more detailed and rigorous treatments can be found in the mixture theory literature, such as Bowen (1980). We write momentum balances for each individual phase as:

$$\begin{aligned} 0 &= \phi_l \rho_l \mathbf{g} + \nabla \cdot (\phi_l \boldsymbol{\tau}_l) - \nabla(\phi_l p_l) + \mathbf{F}_l, \\ 0 &= \phi_s \rho_s \mathbf{g} + \nabla \cdot (\phi_s \boldsymbol{\tau}_s) - \nabla(\phi_s p_s) - \mathbf{F}_s, \end{aligned} \quad (1.10)$$

where $\mathbf{F}_l = \mathbf{F}_s$ are the interphase forces between solid and liquid phases. The interphase forces between the solid and liquid phases are equal and opposite, in accordance with Newton's third law. Neglecting the deviatoric stress in the liquid phase and retaining only the isotropic pressure component, we balance the remaining stress with the viscous drag force and the equilibrium body force. Accordingly, we obtain the following relation for the relative velocity between phases, in alignment with Darcy's law

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \quad (1.11)$$

where k_{ϕ_l} is the matrix permeability, which usually takes the quadratic form as $k_{\phi_l} = k_0 \phi_l^2$.

1.1.4 Compaction

In the context of mantle dynamics, an additional yet unique and essential relationship regulates melt migration. We now revisit the continuous equation (1.6), and consider the two phases separately with the effects of melting incorporated, i.e. ,

$$\begin{aligned}\frac{\partial \phi_l}{\partial t} + \nabla \cdot (\phi_l \mathbf{v}_l) &= \Gamma, \\ \frac{\partial (1 - \phi_l)}{\partial t} + \nabla \cdot ((1 - \phi_l) \mathbf{v}_s) &= -\Gamma,\end{aligned}\tag{1.12}$$

where Γ is the unknown melting rate. We focus on the solid phase equation to get the evolving compaction relationship as

$$\frac{\partial \phi_l}{\partial t} + \mathbf{v}_s \cdot \nabla \phi_l = \Gamma + (1 - \phi_l) \nabla \cdot \mathbf{v}_s,\tag{1.13}$$

where the divergence term $\nabla \cdot \mathbf{v}_s$ is often referred to as the compaction rate. We approximate the static compaction relation as

$$\xi_{\phi_l} \nabla \cdot \mathbf{v}_s = p_l - p_s,\tag{1.14}$$

where ξ_{ϕ_l} is often referred to as bulk viscosity or compaction viscosity. The form of the compaction viscosity ξ_{ϕ_l} has been the subject of extensive discussion. McKenzie (1984) popularized the concept of compaction viscosity and relating the term to the stress balance in a partially molten system. Bercovici et al. (2001) extended this formulation by incorporating surface tension effects. Rabinowicz et al. (2002) further examined compaction behavior in the limit of infinitesimally small porosity. Following the approaches of Hewitt and Fowler (2008) and Katz (2008), we approximate the compaction viscosity ξ_{ϕ_l} using a simplified constitutive relationship,

$$\xi_{\phi_l} = \frac{\mu_s}{\phi_l},\tag{1.15}$$

which highlights the fact that increased melting facilitates the compaction of the solid phase, while in the limit of no melting ($\phi_l = 0$), compaction is entirely eliminated.

1.1.5 Full Static System

By combining equations (1.7), (1.9), (1.11) and (1.14), We finally arrive at the full static flow mechanical system,

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \quad (1.16)$$

$$\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) = 0, \quad (1.17)$$

$$\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) + \bar{\rho} \mathbf{g} = 0, \quad (1.18)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \phi_l(p_l - p_s) = 0, \quad (1.19)$$

where the deviatoric stress of the solid is

$$\boldsymbol{\tau}(\mathbf{v}_s) = 2\mu_s \phi_s (\mathcal{D}\mathbf{v}_s - \frac{1}{3} \nabla \cdot \mathbf{v}_s \mathbf{I}), \quad (1.20)$$

and $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T)$ is the symmetric gradient. Moreover, we adjust the relative permeability $k_{\phi_l} = k_0 \phi_l^{2+2\Theta}$ with constant $\Theta \in [0, 1/2]$. Here μ_s and μ_l are dynamic viscosities for solid and liquid respectively. This complete static mechanical system relates the velocities and pressures of the solid and liquid phases to a given porosity distribution, thereby laying the foundation for subsequent computations.

1.1.6 Scaled Formulation

The Darcy equation mathematically degenerates in the regions where the porosity is zero. To handle this issue, we use the formulation developed previously by Arbogast et al. (2017). We first define pressure potentials as

$$q_l = p_l - \rho_l g z \quad \text{and} \quad q_s = p_s - \rho_l g z, \quad (1.21)$$

where $g = |\mathbf{g}|$ is the magnitude of gravity acceleration and z is the depth, such that $\mathbf{g} = g \nabla z$. Note that ρ_l is used to define both potentials, so that with $q = \bar{q} = \bar{p} - \rho_l g z$,

$$p_l - p_s = q_l - q_s = \frac{1}{1 - \phi_l}(q_l - q).$$

In the mechanical system, the subscript on ϕ_l will be omitted. and we will simply write it as ϕ . Accordingly, the solid phase fraction will be expressed as $1 - \phi$, emphasizing the constraint that the two phase fractions sum to one. Thus

$$\phi \mathbf{v}_r + \frac{k_0 \phi^{2+2\Theta}}{\mu_l} \nabla q_l = 0, \quad (1.22)$$

$$\mu_s \nabla \cdot (\phi \mathbf{v}_r) + \frac{\phi}{1 - \phi} (q_l - q) = 0, \quad (1.23)$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1 - \phi) \rho_r \mathbf{g}, \quad (1.24)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi}{1 - \phi} (q_l - q) = 0, \quad (1.25)$$

where the relative velocity $\mathbf{v}_r = \mathbf{v}_l - \mathbf{v}_s$ and the density difference $\rho_r = \rho_s - \rho_l$. Scaled variables are also defined,

$$\tilde{\mathbf{v}}_r = \phi^{-\Theta} \mathbf{v}_r \quad \text{and} \quad \tilde{q}_l = \phi^{1/2} q_l, \quad (1.26)$$

and the problem is reformulated as

$$\tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_f} \nabla (\phi^{-1/2} \tilde{q}_l) = 0, \quad (1.27)$$

$$\mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1 - \phi} (\tilde{q}_l - \phi^{1/2} q) = 0, \quad (1.28)$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1 - \phi) \rho_r \mathbf{g}, \quad (1.29)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1 - \phi} (\tilde{q}_l - \phi^{1/2} q) = 0, \quad (1.30)$$

where the new scaled relative velocity maintains numerical stability in the zero porosity regions.

1.2 Transport of Chemical Species and Energy

With the velocities of solid and liquid phases determined, we are able to evolve our system forward in time. In our system, chemical species and energy are being moved along with both the solid and the liquid phase.

1.2.1 Transport of Chemical Species

The realistic chemical compositions of mantle and magma are highly complicated and very hard to trace at the same time. Therefore, we present a simple two-species two-phase model. In this framework, we also neglect detailed description of the chemical reaction process and exclude the effect of the change of Gibbs free energies. Similar discussion can be found in the literature like Spiegelman and Elliott (1993), Hewitt and Fowler (2008), Katz (2008) and Hesse et al. (2022). Both species can exist in either the solid or liquid phase. They are assumed to be immiscible in the solid state but fully miscible in the liquid phase. By again extending the Boussinesq approximation, we consider a homogenous density ρ for both solid phase and liquid phase. Figure 1.1 illustrates the coexistence of species and phases in representative volume element. Within the representative volume element V , $\gamma_{i,j}$ represents one

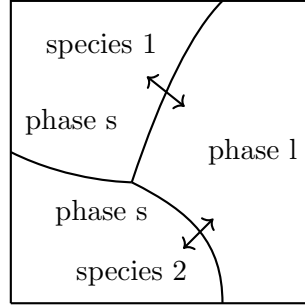


Figure 1.1: Phases and species in an representative volume element (RVE).

quantity of the species i present in phase j , where $i \in 1, 2$ and $j \in \{s, l\}$. The mass fraction $c_{i,j}$ of a given species i in phase j is also regarded as a concentration. We can write transport equations for an individual species i , assuming a pseudo melting term Γ representing the transition between solid phase and liquid phase as

$$\frac{\partial}{\partial t}(\phi_l \rho_l c_{i,l}) + \nabla \cdot (\phi_l \rho_l c_{i,l} \mathbf{v}_l - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}) = \Gamma, \quad (1.31)$$

$$\frac{\partial}{\partial t}(\phi_s \rho_s c_{i,s}) + \nabla \cdot (\phi_s \rho_s c_{i,s} \mathbf{v}_s - \phi_s \rho_s \mathcal{D}_{i,s} \nabla c_{i,s}) = -\Gamma, \quad (1.32)$$

where $\mathcal{D}_{i,l}$ and $\mathcal{D}_{i,s}$ are the diffusion coefficients of component i in the liquid and the solid phase respectively. The diffusion term will be dropped in the solid phase. By

summing up equations (1.31) and (1.32), we obtain an advection-diffusion equation as

$$\frac{\partial \overline{\rho c_i}}{\partial t} + \nabla \cdot (\overline{\rho c_i \mathbf{v}} - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}) = 0, \quad (1.33)$$

where the phase averaged quantities are $\overline{\rho c_i} = \sum_{j \in \{s,l\}} \phi_j \rho_j c_{i,j}$ and $\overline{\rho c_i \mathbf{v}} = \sum_{j \in \{s,l\}} \phi_j \rho_j c_{i,j} \mathbf{v}_j$. The subscript i will be omitted, as only one chemical species needs to be explicitly computed in this two-species system. By extending the Boussinesq approximation such that $\rho_l = \rho_s = \bar{\rho}$, the species transport equation can be simplified to the form

$$\frac{\partial \bar{c}}{\partial t} + \nabla \cdot (\bar{c} \mathbf{v} - \phi_l \mathcal{D}_l \nabla c_l) = 0. \quad (1.34)$$

1.2.2 Effective Velocity

The concept of effective velocity has long been employed in chemistry, particularly in the context of particle transport. In mantle dynamics, Spiegelman (1996) applied effective velocity for equilibrium transport of concentration within the liquid phases. We define a corresponding effective velocity for the transport of bulk composition. To begin, we define the partition coefficient as $D_i = c_{i,s}/c_{i,l}$. The concentration of species i in the liquid and solid phase can be expressed in terms of the bulk composition as follows:

$$(\phi_l + \phi_s D_i) c_{i,l} = \bar{c}_i, \quad (1.35)$$

$$(\phi_l / D_i + \phi_s) c_{i,s} = \bar{c}_i. \quad (1.36)$$

By expressing $c_{i,l}$ and $c_{i,s}$ with bulk composition given by equations (1.35) and (1.36), we can rewrite the bulk velocity as

$$\bar{c}_i \mathbf{v} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l} = \left(\frac{\phi_l \mathbf{v}_l + D_i \phi_s \mathbf{v}_s}{\phi_l + \phi_s D_i} \right) \bar{c}_i. \quad (1.37)$$

We get the proper expression for the effective velocity, denoted as

$$\mathbf{v}_{\text{eff}} = \frac{\phi_l \mathbf{v}_l + D_i \phi_s \mathbf{v}_s}{\phi_l + \phi_s D_i}, \quad (1.38)$$

which is the appropriate velocity at which the bulk composition should advect. In the absence of melting, where $\phi_l = 0$, we retrieve $\mathbf{v}_{\text{eff}} = \mathbf{v}_s$. Conversely, in the case of complete melting, where $\phi_l = 1$, we obtain $\mathbf{v}_{\text{eff}} = \mathbf{v}_l$. Therefore, the definition of effective velocity is well defined in those two limiting scenarios. We rewrite the transport equation (1.34) with effective velocity (1.38) as

$$\frac{\partial C}{\partial t} = \nabla \cdot (\mathbf{v}_{\text{eff}} C - \phi_l \mathcal{D}_l \nabla c_l) = 0, \quad (1.39)$$

where $C = \bar{c}$ for simplicity.

1.2.3 Transport of Energy

According to the first law of thermodynamics, we write the energy balance equation in the phase-averaged internal energy form as

$$\frac{\partial \overline{\rho u}}{\partial t} + \nabla \cdot \overline{\rho u \mathbf{v}} = \overline{\rho \mathcal{Q}} + \nabla \cdot \overline{K \nabla T} + \nabla \cdot \overline{\boldsymbol{\sigma} \cdot \mathbf{v}} + \overline{\rho \mathbf{v}} \cdot \mathbf{g}, \quad (1.40)$$

where $\overline{K} = \phi_l K_l + \phi_s K_s$ is the phase averaged thermal conductivity. On the left-hand side, the equation represents the time-dependent rate of change of internal energy of a two-phase mixture as it moves with the flow. On the right-hand side, it includes the heat source term $\mathcal{Q}$, conductive heat transfer following Fourier's law, mechanical work done by mixture stress, and work against gravity respectively.

In a thermochemical equilibrium scenario, we replace internal energy u with enthalpy h . The specific enthalpy is defined as

$$h = u + PV/\rho, \quad (1.41)$$

We formulate the enthalpy equation by neglecting the external heating/cooling term $\mathcal{Q}$ and dissipative heating following the discussion presented by Katz (2008),

$$\frac{\partial \overline{\rho h}}{\partial t} + \nabla \cdot \overline{\rho h \mathbf{v}} = \frac{\partial P}{\partial t} + \overline{v} \cdot \nabla P + \nabla \cdot \overline{K \nabla T}. \quad (1.42)$$

1.2.4 Temperature Formulation

The specific relates to temperature directly with specific heat capacity as

$$h - h_0 = c_p(T - T_0), \quad (1.43)$$

where the reference values h_0 and T_0 are set as zero for simplicity. Another important physical attribute in the context of melting is latent heat L . We write specific enthalpy in the solid and the liquid phase separately as

$$\begin{aligned} h_s &= c_p T, \\ h_l &= c_p T + L. \end{aligned} \quad (1.44)$$

We rewrite the enthalpy equation with latent heat and temperature as

$$\frac{\partial \overline{\rho h}}{\partial t} + \rho c_p \nabla \cdot (\overline{\mathbf{v}} T) = \rho L \nabla \cdot (\phi_s \mathbf{v}_s) + \overline{K} \nabla^2 T + \rho \alpha_p T \overline{\mathbf{v}} \cdot \mathbf{g}, \quad (1.45)$$

where one specific heat capacity c_p is used for both components in solid and liquid phase. The last term including gravitational acceleration has previously been referred to as adiabatic cooling or decompression cooling by McKenzie and Bickle (1988). Assuming no density difference between the solid and the liquid phase, we divide ρc_p on both sides of the equation to obtain a normalized form as

$$\frac{\partial \overline{H}}{\partial t} + \nabla \cdot (\overline{\mathbf{v}} T) = \frac{L}{c_p} \nabla \cdot (\phi_s \mathbf{v}_s) + \kappa \nabla^2 T + \frac{\alpha_p}{c_p} T \overline{\mathbf{v}} \cdot \mathbf{g}, \quad (1.46)$$

where $\overline{H} = \overline{h}/c_p$ is the bulk specific enthalpy and $\kappa = \overline{K}/\rho c_p$ is the bulk heat diffusivity.

1.3 Eutectic Phase Behavior

With the transport equations defined, we can now proceed to redistribute chemical species and energy within the system. The next step is to determine the thermodynamic state of the system, assuming thermodynamic equilibrium is attained at each time step. Recognizing that mid-ocean ridges form in regions where temperatures are generally lower, eutectic melting, where the presence of a secondary species

significantly lowers the melting temperature, provides a suitable model for describing the melting behavior in these areas.

The binary eutectic system has been widely applied in studies involving chemical equilibrium, with detailed description can be found in textbooks such as Lupis (1983). In the context of mantle dynamics, eutectic melting was being incorporated by Ribe (1985). Moreover, eutectic phase behavior has been applied to solidification of alloys by Wheeler et al. (1996). More recently, binary eutectic melting has also been applied to water-salt-brine systems in glaciers Hesse et al. (2022).

We assume the two primary components in our partially molten system are olivine and orthopyroxene(opx), which are highly miscible in the liquid phase, but immiscible in the solid phase. Olivine is considered the dominant component, while orthopyroxene represents the minor component. Assuming thermodynamic equilibrium is attained at each time step, we can determine the phase fraction of liquid, the volume fraction of each solid components and the temperature using a binary eutectic phase diagram with transported enthalpy and bulk composition.

A typical eutectic phase diagram is shown in Figure 1.2. There are two key

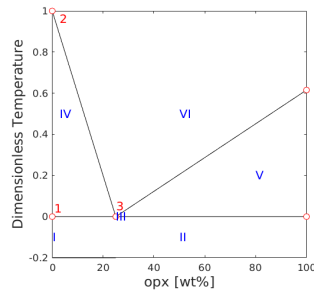


Figure 1.2: Eutectic Phase Diagram of a Olivine-Orthopyroxene system.

concepts about the point 3 on the diagram:

- Eutectic melting point or eutectic temperature T_e , is the lowest temperature a eutectic system starts melting. The melting happens at eutectic temperature

is called eutectic melting. The temperature of the system is fixed as eutectic temperature when eutectic melting happens.

- Eutectic composition C_e is a fixed opx-olivine concentration ratio of the liquid produced by eutectic melting.

1.3.1 Clausius-Clapeyron Relation

In geological contexts, pressure can undergo significant variations during dynamic events such as mantle upwelling. As a result, the influence of pressure on the melting point cannot be neglected. To account for this, we approximate the solidus melting temperature T_m at a given pressure using the approximated Clausius-Clapeyron relationship. We also extend such approximation to eutectic temperature T_e as

$$\begin{aligned} T_m &= T_{m0} + \nu^{-1}(P - P_0), \\ T_e &= T_{e0} + \nu^{-1}(P - P_0), \end{aligned} \tag{1.47}$$

where ν is the Clausius-Clapeyron constant, and P_0 is the reference pressure usually set to be 0. The melting point T_{m0} and eutectic point T_{e0} refer to their respective values at standard atmospheric pressure.

1.3.2 Some Assumptions

In a eutectic system, the current thermodynamic state of the material can be identified using the bulk mass fraction of opx (orthopyroxene) and bulk enthalpy of the system. Our bulk mass fraction will refer to orthopyroxene(opx) as we consider a binary system involving only two components. We begin by introducing some assumptions regarding densities and other material properties.

- We assume no density difference between solid phase olivine and opx, i.e., $\rho_{\text{opx}} = \rho_{\text{olivine}}$.
- We assume no density change during phase transition (extended Boussinesq approximation), i.e., $\rho_l = \rho_s$.

- We assume constant and uniform material thermodynamics properties like specific heat capacity c_p and thermal expansion coefficient α_p .

Following the assumptions, we relate the bulk composition with previously defined phase averaged concentration (1.39) as

$$\frac{\phi_l \rho_l c_l + \phi_s \rho_s c_s}{\phi_l \rho_l + \phi_s \rho_s} = \phi_l c_l + \phi_s c_s = \phi_2 + \phi_l c_l = C. \quad (1.48)$$

We also express bulk specific enthalpy with temperature and latent heat

$$\bar{H} = \phi_l h_l / c_p + \phi_s h_s / c_p + \phi_l L / c_p = \phi_l T + \phi_s T + \phi_l L / c_p = T + \phi_l L / c_p. \quad (1.49)$$

According to equations (1.47), the difference between melting point and eutectic point, defined as $\Delta T = T_m - T_e = T_{m0} - T_{e0}$, which remains constant regardless of pressure change. At any given pressure, we proceed by dividing this ΔT to get a dimensionless formulation as

$$\frac{\bar{H}}{\Delta T} = \frac{T}{\Delta T} + \frac{\phi_l L}{c_p \Delta T}. \quad (1.50)$$

Let $\check{L} := \frac{L}{c_p \Delta T}$, $H := \frac{\bar{H}}{\Delta T}$ and $\check{T} := \frac{T}{\Delta T}$. We rewrite the enthalpy temperature relationship as

$$H = \check{T} + \phi_l \check{L}. \quad (1.51)$$

1.3.3 Phase Diagram

Binary eutectic melting is usually highly-nonlinear. For simplicity, we adopt a geometric parameterization of the binary eutectic diagram that can be analytically evaluated. We implement a linear eutectic phase boundary to distinguish between different phase regions within the diagram.

We begin with a brief description of eutectic phase diagram illustrated in Figures 1.2. The eutectic composition is set as $C_e \approx 0.25$. For illustration only, we plot bulk composition against a renormalized dimensionless temperature $\check{T} - T_e / \Delta T$, where $\check{T}_m - T_e / \Delta T = 1$ and $\check{T}_e - T_e / \Delta T = 0$.

We identify the following 6 regions with different phase assemblages:

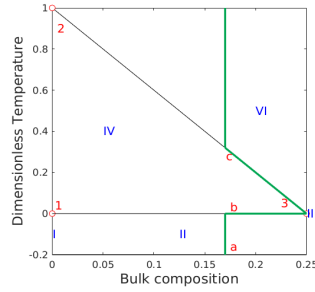


Figure 1.3

Figure 1.4: Phase diagram when composition of opx smaller than eutectic composition.

- I Single-phase sub-solidus : Olivine
- II Two-phase sub-solidus : Olivine + Orthopyroxene
- III Three-phase eutectic : Olivine + Orthopyroxene + melt
- IV Two-phase super-eutectic : Olivine + melt
- V Two-phase super-eutectic : Orthopyroxene + melt
- VI Single-phase super-liquidus : melt

In partial melting system, we assume the bulk composition of orthopyroxene (opx) remains below the eutectic composition. Consequently, we focus in the part of phase diagram 1.2, where bulk composition is less than eutectic composition, which is illustrated in Figure 1.3.

The melting in eutectic system is illustrated with the green line in Figure 1.3. Now imagining a system containing 17% opx starts melting. Starting with all solid phase marked as point *a*, the system temperature increase linearly in response to the enthalpy rise. When the temperature reaches the eutectic temperature, marked as point *b*, both components of the system starts melting with a fixed ratio of 25% opx and 75% olivine, which is the eutectic composition. Once all opx is gone, the system leaves the eutectic points and climbs up along the line from point 3 to point *c*. After

all solid exhausted, the temperature of the system will continuous to rise linearly in response to enthalpy input. It is worth noting that we can reverse the process to obtain a path of equilibrium crystallization.

1.3.4 Phase Split Calculation

A detailed description of the phase-splitting calculations is provided for Regions I, II, III, and IV in this chapter. We draw a Enthalpy-Composition diagram including the effect of latent heat in Figure 1.5. Again, we plot against the renormalized enthalpy $H - T_e/\Delta T$ for simplicity. The individual concentration and temperature are determined using equations (1.48) and (1.51). We denote the volume fraction of olivine as ϕ_1 and the volume fraction of orthopyroxene as ϕ_2 , such that $\phi_1 + \phi_2 = \phi_s$.

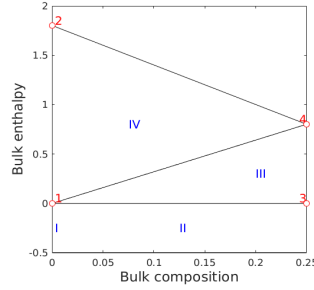


Figure 1.5: Enthalpy-Composition Diagram

Region I. Pure olivine in solid phase. The melting of pure olivine does not follow eutectic melting path. The conditions to be in region I are

$$C = 0, \quad H \leq \check{T}_m, \quad (1.52)$$

and we can determine phase fractions and temperature as

$$\phi_1 = 1, \quad \phi_2 = \phi_l = 0 \quad \text{and} \quad \check{T} = H. \quad (1.53)$$

We can determine partial derivatives of temperature with respect to bulk enthalpy and bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 1 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \quad (1.54)$$

Region II. Sub-solidus: two components coexist as solid phase. In this phase region, temperature hasn't reached eutectic melting temperature. The conditions to be in region II are

$$C \neq 0, \quad H \leq \check{T}_e, \quad (1.55)$$

and we can determine phase fractions and temperature as

$$\phi_2 = C, \quad \phi_1 = 1 - \phi_2, \quad \phi_l = 0 \quad \text{and} \quad \check{T} = H. \quad (1.56)$$

We can determine partial derivatives of temperature with respect to bulk enthalpy and bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 1 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \quad (1.57)$$

Region III. Two-phase eutectic: The system starts melting at eutectic temperature with a fixed composition ratio. Both olivine and orthopyroxene in solid state and melt coexist at the same time. Temperature remains eutectic point until opx exhausts. Concentration of opx in the melt is also fixed equal to eutectic composition. The conditions to be in region III are

$$C \leq C_e, \quad \check{T}_e < H \leq \text{line}_{1-4}, \quad (1.58)$$

where line_{1-4} is defined as $H(C) = \check{T}_e + \check{L}C/C_e$, and we can determine phase fractions and current temperature,

$$\phi_l = \frac{H - \check{T}_e}{\check{L}}, \quad \phi_2 = C - \phi_l C_e, \quad \phi_1 = 1 - \phi_l - \phi_2 \quad \text{and} \quad \check{T} = \check{T}_e. \quad (1.59)$$

We can determine partial derivatives of temperature with respect to bulk enthalpy and bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 0 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \quad (1.60)$$

Region IV. Super-eutectic two phase region. All solid phase opx has been exhausted. The concentration of opx in liquid c_l is now approximated with a linear function. The conditions to be in region IV are

$$C \leq C_e, \quad \text{line}_{1-4} < H \leq \text{line}_{2-4}, \quad (1.61)$$

where line_{2-4} is defined as $H(C) = H_m - c_l/C_e$, and we can determine phase fractions as

$$\phi_2 = 0, \quad \phi_l = \frac{C}{c_l(\check{T})} \quad \text{and} \quad \phi_1 = 1 - \phi_l. \quad (1.62)$$

The corresponding temperature can be determined by solving the following equation

$$F(\check{T}) = H - \check{T} - \frac{C}{c_l(\check{T})}\check{L} = 0. \quad (1.63)$$

Let $c_l = C_e(\check{T}_m - \check{T})$, we can obtain a quadratic equation about temperature as

$$\check{T}^2 - (H + \check{T}_m)\check{T} + \check{T}_m H - CL/C_e = 0, \quad (1.64)$$

which can be solved with standard quadratic root finding formula. We omit one of the roots that exceeding melting temperature,

$$\begin{aligned} \check{T} &= \frac{H + \check{T}_m - \sqrt{(H + \check{T}_m)^2 - 4(\check{T}_m H - C\check{L}/C_e)}}{2} \\ &= \frac{H + \check{T}_m - \sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}{2}. \end{aligned} \quad (1.65)$$

We can determine partial derivatives of temperature with respect to bulk enthalpy and bulk composition as

$$\begin{aligned} \frac{\partial \check{T}}{\partial H} &= \frac{1}{2} \left(1 - \frac{H - \check{T}_m}{\sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}} \right) \\ \frac{\partial \check{T}}{\partial C} &= - \frac{\check{L}/C_e}{\sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}. \end{aligned} \quad (1.66)$$

1.4 Nondimensionalization

Compaction and melting are two most important aspects in mantle dynamics. We nondimensionlize the scaled formulation with characteristic length scale related to compaction length and scaled characteristic Darcy speed of the melt,

$$l_0 = \left(\frac{k_0 \mu_s}{\mu_l} \right)^{1/2} \quad \text{and} \quad u_0 = \frac{k_0 \rho_r g}{\mu_l}. \quad (1.67)$$

We define corresponding characteristic time scale as $t_0 = l_0/u_0$, and characteristic pressure scale as $p_0 = \rho g l_0$. Now we can write a full set of dimensionless equations. A set of nondimensional variables are defined as

$$\check{\mathbf{v}}_r = \frac{\mathbf{v}_r}{u_0}, \quad \check{\mathbf{v}}_s = \frac{\mathbf{v}_s}{u_0}, \quad \check{q} = \frac{q}{p_0}, \quad \check{q}_l = \frac{q_l}{p_0}. \quad (1.68)$$

The resulting dimensionless mechanics and transport equations are

$$\check{\mathbf{v}}_r + \phi^{1+\Theta} \nabla (\phi^{-1/2} \check{q}_l) = 0, \quad (1.69)$$

$$\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_r) + \frac{1}{1-\phi} (\check{q}_l - \phi^{1/2} \check{q}) = 0, \quad (1.70)$$

$$\nabla \check{q} - \nabla \cdot \check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi) \mathbf{n}, \quad (1.71)$$

$$\nabla \cdot \check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi} (\check{q}_l - \phi^{1/2} \check{q}) = 0, \quad (1.72)$$

where dimensionless deviatoric stress of the solid phase is $\check{\tau} = 2(1-\phi)(\frac{1}{2}(\nabla \check{\mathbf{v}}_s + \nabla \check{\mathbf{v}}_s^T) - 1/3 \nabla \cdot \check{\mathbf{v}}_s \mathbf{I})$, which maintains the same form as (1.20). $\mathbf{n}$ is the unit direction vector pointing downwards. Now we rewrite transport equations with dimensionless variables as

$$\begin{aligned} \frac{\partial C}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}}_{\text{eff}} C - \frac{\phi \mathcal{D}_l}{u_0 l_0} \nabla c_l) &= 0, \\ \frac{\partial H}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}} \check{T}) - \check{L} \nabla \cdot \phi_s \check{\mathbf{v}}_s - \frac{\kappa}{u_0 l_0} \nabla^2 \check{T} - \frac{\alpha_\rho l_0}{c_p} \check{T} \check{\mathbf{v}} \cdot \mathbf{g} &= 0. \end{aligned} \quad (1.73)$$

Parameters used in the flow mechanics and thermodynamic computations are listed in Table 1.1,

Quantity	Value	Units
ρ	3000	$kg \cdot m^{-3}$
ρ_r	600	$kg \cdot m^{-3}$
α_ρ	3×10^{-5}	K^{-1}
c_p	1200	$J \cdot kg^{-1} \cdot K^{-1}$
L	5×10^5	$J \cdot kg^{-1}$
μ_l	1	Pascal $\cdot s$
μ_s	10^{19}	Pascal $\cdot s$
k_0	10^{-8}	m^2
T_{e0}	1480	K
T_{m0}	2000	K
ν	6.5×10^6	Pascal $\cdot K^{-1}$

Table 1.1: Flow Mechanics and Thermaldynamics Parameters

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